Example 2.4

The analysis of a network begins with the elements it contains and the way they are joined. Each branch carries a current, and each node has a potential measured against a common reference. Kirchhoff's laws relate these quantities: the currents entering a node sum to zero, and the voltages around any closed loop sum to zero. With Ohm's law for every resistor, these relations give as many independent equations as there are unknowns, so the circuit can be solved completely. In practice the equations are written so that the number of unknowns stays small, either by choosing loop currents or node voltages as the variables. The analysis of a network begins with the elements it contains and the way they are joined. Each branch carries a current, and each node has a potential measured against a common reference. Kirchhoff's laws relate these quantities: the currents entering a node sum to zero, and the voltages around any closed loop sum to zero. With Ohm's law for every resistor, these relations give as

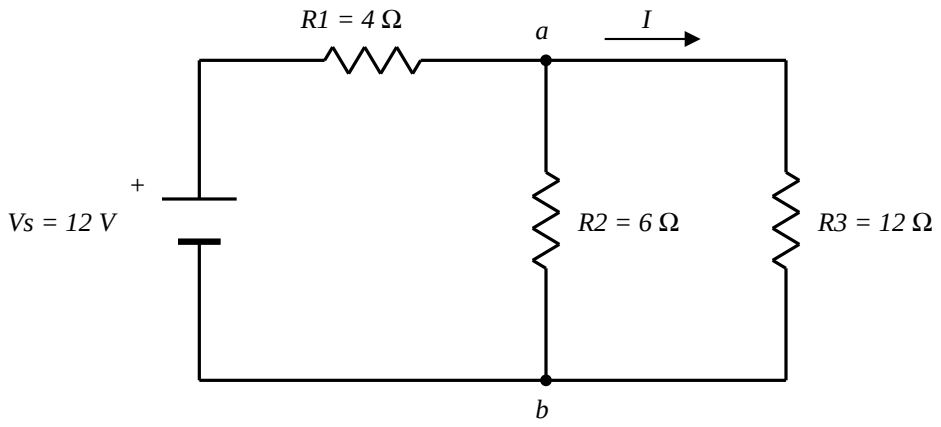

Figure 2.4 A series–parallel circuit. The source drives R1 in series with the parallel pair R2 and R3; the current I divides at node a.

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$$Veq = R1 + R2\,R3 / (R2 + R3) \tag{2.3}$$

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